

Host (CPU)

Model binary
.bin (FP32+INT8)

Memory-mapped I/O
Zero-copy

FP32
Embed

INT8
W+scale

Tokenizer
BPE (Llama 3.2)

Sampler
top-p / greedy

Device (GPU): Pre-allocated VRAM Pool (2054 MB)

VRAM pool contents

FP32 Emb

INT8 W

KV cache

Activ.

RMSNorm

INT8 GEMV (Q/K/V)
Opt 1 (S3) + Opt 4 (S6)

RoPE + KV update

Fused attention
Opt 3 (S5): 35 → 1 launch

INT8 GEMV (output)
Opt 1 (S3) + Opt 4 (S6)

Fused Add+RMSNorm
Opt 2 (S4)

INT8 GEMV (FFN gate/up/down)
Opt 1 (S3) + Opt 4 (S6)

SiLU + residual

FP32 logits
vocab=128,256

token IDs

16 layers

Six-Stage Roadmap

S1 FP32 | 14.9 tok/s

S2 Naive INT8 | 16.1

S3 Warp-Opt | **31.8 (2.1 ×)**

S4 +Fused Ops | 30.0

S5 +Flash Attn | 35.2

S6 dp4a W8A8 | **44.0 (2.95 ×)**

PPL 13.36 | +1.2% vs FP16

Validation

Cross-architecture (1B):

RTX 4060 (Ada 8.9)

RTX 3090 (Ampere 8.6)

RTX 2080 Ti (Turing 7.5)

Cross-scale:

1B: 2.95 × cumulative

3B: 4.36 × cumulative

Energy (S5 → S6):

+31% tok/J

same 76.7 W